

AKOL PAUL

SOFTWARE ENGINEER

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PROFILE SUMMARY

Software Engineer with hands-on experience delivering production systems across fintech, real-time data platforms, humanitarian technology, and AI/ML. Proficient in full-stack development using Django, Laravel, React, and React Native, with strong command of RESTful and GraphQL APIs, microservices architecture, and OOP principles. Experienced in secure, scalable deployments using Docker, CI/CD pipelines, PostgreSQL, MySQL, and Firebase. Internship-trained at NSSF Uganda on enterprise-level systems. Open to full-time, contract, and remote engagements.

TECHNICAL SKILLS

Languages: Python, Java, PHP, JavaScript, TypeScript

Frameworks & Libraries: Django, Django REST Framework, Laravel, React, React Native, Node.js, Flask

Databases: PostgreSQL, MySQL, Firebase, Redis

DevOps & Cloud: Docker, CI/CD, Railway, Vercel, Git, GitHub

APIs & Architecture: RESTful APIs, GraphQL, WebSockets, Microservices, JWT Authentication

Security: OWASP practices, Role-Based Access Control, Brute-force lockout, Secure session management

Tools: Jira, ClickUp, Postman, Linux

WORK EXPERIENCE

National Social Security Fund (NSSF) Uganda | Intern

June 2025 - August 2025

- Contributed to development and maintenance of enterprise-level software applications.
- Designed and integrated RESTful APIs to support internal systems.
- Worked with PostgreSQL and Firebase for secure data management and real-time features.
- Applied OOP principles in backend development using Java, PHP (Laravel), and Python (Django).
- Collaborated with cross-functional teams using Jira and ClickUp in agile workflows.
- Participated in code reviews and debugging to ensure maintainability and performance.
- Assisted in deploying applications in Dockerized environments with CI/CD pipelines.
- Gained exposure to cybersecurity and network practices for secure system design.

PROJECTS

Project		Stack	Highlight
Afriplay Feed	Sports	Django, React, WebSockets, Railway, Vercel	Live real-time sports reporting platform with role-separated portals for admins, moderators, and reporters
RefuConnect		Django, NLP, fastText, HuggingFace, Google Cloud	Humanitarian feedback platform with real-time NLP pipeline processing multilingual input from Uganda refugee settlements

SALT-KD	Python, Whisper HuggingFace, Colab	large-v2,	Knowledge distillation ASR system compressing Whisper for 7 Ugandan/African languages
8TechBank	Flask, Jinja2, PostgreSQL		Dual-mode banking app production-hardened and intentionally vulnerable build for security research

EDUCATION

Makerere University

Bachelor of Science in Software Engineering

2022 - *Expected 2027*

Bukedea Lifeline S.S

Uganda Advanced Certificate of Education (UACE)

2019 - 2020

Bukedea Lifeline S.S

Uganda Certificate of Education (UCE)

2015 - 2018

LANGUAGES

English (Fluent) | Ateso (Fluent)

REFERENCES

Prof. Tonny Oyana

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